CREDIT PRE-SCREENING ENGINE: LOAN INTAKE OPTIMISATION

Individual Contribution Report

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Abstract

This report documents my contribution to the MUC02 Credit Pre-Screening Engine, focusing on Steps 5.6–5.7: fairness analysis by gender and employment type, examination of potential proxy effects, and preparation of the final credit-policy briefing. The fairness audit used the held-out test set and the validation-selected threshold of 0.95. The model auto-decline rate was 10.64% for female applicants and 9.19% for male applicants. Female applicants also had a higher Gig / Contract employment share (20.45% compared with 11.51%). Within Gig / Contract employment, auto-decline rates were 23.29% for female applicants and 21.13% for male applicants. The final policy evaluation recorded a 1.30% wrong-decline share on the test set and estimated approximately 120.3 analyst hours saved per month under the project's operating assumptions.

I. Introduction and Assigned Responsibility

The MUC02 use case examined whether historical personal-loan application data could support a consistent preliminary screening workflow while retaining human oversight. My assigned responsibility was to complete Steps 5.6–5.7. Step 5.6 involved assessing fairness across gender and employment type and examining whether other model inputs could act as proxies. Step 5.7 involved preparing the final credit-policy briefing, bringing together the analysis, recommendations, operational risks and limitations.

II. Fairness Analysis by Gender

The fairness audit was conducted on the held-out test set. Gender was excluded from the model's predictive features and used only for post-model fairness monitoring. The comparison considered application counts, historical approval rates, model auto-decline rates, median monthly income and the proportion of Gig / Contract applicants.

Measure	Female	Male
Test applications	714	1,686
Historical approval rate	51.12%	53.74%
Model auto-decline rate	10.64%	9.19%
Median monthly income	INR 46,000	INR 56,500
Gig / Contract share	20.45%	11.51%

The female group had a 1.45 percentage-point higher auto-decline rate than the male group. It also had lower median monthly income and a larger Gig / Contract share. These are descriptive differences that warrant monitoring; they do not, by themselves, prove discrimination or establish the cause of the difference.

III. Fairness Analysis Within Employment Type

To examine whether the overall gender pattern was associated with employment composition, I compared historical approval and model auto-decline rates separately within each employment category. All gender-by-employment groups contained at least 30 test applications, meeting the notebook's minimum group-size check.

Employment type	Gender	Test n	Historical approval	Auto-decline
Business Owner	Female	60	48.33%	10.00%
Business Owner	Male	297	51.52%	10.77%
Gig / Contract	Female	146	35.62%	23.29%
Gig / Contract	Male	194	34.02%	21.13%
Salaried	Female	409	56.97%	5.87%
Salaried	Male	998	58.72%	6.81%
Self-Employed Professional	Female	99	51.52%	12.12%
Self-Employed Professional	Male	197	51.27%	7.11%

Gig / Contract applicants had the highest auto-decline rates for both genders. The female rate was also higher in the Self-Employed Professional category, while the male rate was slightly higher among Business Owners and Salaried applicants. Therefore, the pattern was not identical across all employment categories.

IV. Potential Proxy Effects

Excluding gender from the model inputs does not automatically eliminate proxy effects. Variables such as employment type, monthly income and affordability measures may be associated with group-level differences and can influence model outcomes. In the test data, female applicants had a larger Gig / Contract share and lower median monthly income. These relationships make subgroup monitoring important even when gender is not directly used for prediction. This analysis is descriptive and does not establish causality or prove that a feature is an impermissible proxy.

V. Final Credit-Policy Briefing

The validation process selected a threshold of 0.95 as the lowest tested threshold meeting the project's maximum 5% wrong-decline constraint. At 0.95, the validation wrong-decline share was 2.53%. The threshold was then evaluated separately on the held-out test set. There, 231 of 2,400 applications (9.62%) entered the auto-decline lane; 3 of these were historically approved, giving a test wrong-decline share of 1.30%. The policy captured 20.19% of historical declines in the test set.

Policy measure	Result
Validation-selected threshold	0.95
Test applications auto-declined	231 of 2,400 (9.62%)
Wrong-decline share on test set	1.30% (3 historically approved cases)
Estimated monthly applications routed away from review	Approximately 289
Estimated analyst time saved	Approximately 120.3 hours per month

The monthly estimate assumes approximately 3,000 applications and 25 minutes of manual review per application. It is a projection based on the project assumptions, not a measured result from live deployment.

VI. Recommendations, Risks and Limitations

Recommendations: Treat the model as a controlled decision-support pilot rather than a final lending decision-maker; keep New-to-Credit (NTC) applicants in manual review; monitor approval and auto-decline rates by gender and employment type; investigate potential proxy effects; and maintain human review and appeal procedures.

Risks and limitations: The target represents historical analyst decisions, not actual repayment or default. Repayment outcomes are unavailable in the current dataset, and out-of-time validation has not been completed. Human-review arrangements, calibration monitoring and drift monitoring require operational approval. The fairness comparisons are descriptive and should not be interpreted as proof of discrimination or causality.

VII. Conclusion

My contribution assessed the fairness of the proposed credit pre-screening policy across gender and employment type, examined potential proxy effects, and translated the findings into recommendations and safeguards. The analysis identified differences in overall auto-decline rates and highlighted elevated auto-decline rates among Gig / Contract applicants of both genders. Although the selected threshold produced a 1.30% wrong-decline share on the held-out test set and an estimated 120.3 analyst hours saved per month, these results must be interpreted within the limitations of historical-decision labels. Continued fairness monitoring, NTC manual review, human oversight, repayment-outcome validation and out-of-time testing are necessary before real-world deployment.

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